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Figure 1: Schematic diagrams of the four CRISPR-Cas9 systems. (A) VOM1271T system with RPVL4, RPVB4, and VOM1271T. (B) SIC-1200-5A system with RPVLB, RPVB8, and SIC-1200-5A. (C) VOM1271T system with RPVLB, RPVB8, and VOM1271T. (D) SIC-1200-5A system with RPVLB, RPVB8, and SIC-1200-5A. Each diagram shows the arrangement of genes (RPVLA, RPVLB, RPVB4, RPVB8), CRISPR arrays (RPVLA, RPVLB, RPVB4, RPVB8), and the Cas9 gene (VOM1271T, SIC-1200-5A).

Figure 10 consists of three schematic diagrams labeled (a), (b), and (c), representing the main power stage, auxiliary power stage, and output filter stage, respectively.

(a) Main power stage: This diagram shows a half-bridge consisting of two MOSFETs (M1, M2) and two anti-parallel diodes (D1, D2). The MOSFETs are driven by gate drivers (G1, G2) with gate resistors (R1, R2) and gate-to-source capacitors (C1, C2). The MOSFETs are connected to a series combination of an inductor (L1) and a capacitor (C1). The MOSFETs are labeled with their part numbers (M1, M2) and gate-to-source capacitors (C1, C2). The MOSFETs are connected to a series combination of an inductor (L1) and a capacitor (C1). The MOSFETs are labeled with their part numbers (M1, M2) and gate-to-source capacitors (C1, C2).

(b) Auxiliary power stage: This diagram shows a half-bridge consisting of two MOSFETs (M3, M4) and two anti-parallel diodes (D3, D4). The MOSFETs are driven by gate drivers (G3, G4) with gate resistors (R3, R4) and gate-to-source capacitors (C3, C4). The MOSFETs are connected to a series combination of an inductor (L2) and a capacitor (C2). The MOSFETs are labeled with their part numbers (M3, M4) and gate-to-source capacitors (C3, C4). The MOSFETs are connected to a series combination of an inductor (L2) and a capacitor (C2). The MOSFETs are labeled with their part numbers (M3, M4) and gate-to-source capacitors (C3, C4).

(c) Output filter stage: This diagram shows a half-bridge consisting of two MOSFETs (M5, M6) and two anti-parallel diodes (D5, D6). The MOSFETs are driven by gate drivers (G5, G6) with gate resistors (R5, R6) and gate-to-source capacitors (C5, C6). The MOSFETs are connected to a series combination of an inductor (L3) and a capacitor (C3). The MOSFETs are labeled with their part numbers (M5, M6) and gate-to-source capacitors (C5, C6). The MOSFETs are connected to a series combination of an inductor (L3) and a capacitor (C3). The MOSFETs are labeled with their part numbers (M5, M6) and gate-to-source capacitors (C5, C6).

Figure 1 shows schematic diagrams of various circuit components:

- (a) **JOLTIP** and **STUD-M8**: A JOLTIP component is connected to a STUD-M8 component.
- (b) **SHUNT-MANG**: A complex circuit involving a **PSH0** component, a **US200-200V** component, and a **US200-200V** component. It also includes a **US200-200V** component and a **US200-200V** component.
- (c) **NSH1200-D5WR**: A circuit involving a **PSH1** component, a **US200-200V** component, and a **US200-200V** component. It also includes a **US200-200V** component and a **US200-200V** component.
- (d) **ISOSV-RFC-6K**: A circuit involving a **PSH1** component, a **US200-200V** component, and a **US200-200V** component. It also includes a **US200-200V** component and a **US200-200V** component.
- (e) **JOLTIN** and **STUD-M8**: A JOLTIN component is connected to a STUD-M8 component.
- (f) **COF1** and **COF2**: Two components, COF1 and COF2, are shown. COF1 is connected to a **US200-200V** component. COF2 is connected to a **US200-200V** component.
- (g) **CYO3** and **CYO4**: Two components, CYO3 and CYO4, are shown. CYO3 is connected to a **US200-200V** component. CYO4 is connected to a **US200-200V** component.

The diagram illustrates the genetic constructs used in the study. At the top, the RTLLCP construct is shown with a VSP3 promoter driving a 10xK enhancer, which in turn drives the T-LLCP gene. Below this, the RTXFRFP construct is shown with a VSP3 promoter driving a 10xK enhancer, which drives the TTXFRFP gene. The JTXFRFP construct is shown with a T-LLCP promoter driving a PH-2 enhancer, which drives the JTXFRFP gene. The T-LLCP promoter is flanked by a 10xK enhancer and a T-LLCP gene. The TTXFRFP promoter is flanked by a 10xK enhancer and a TTXFRFP gene. The JTXFRFP promoter is flanked by a 10xK enhancer and a JTXFRFP gene. The PH-2 enhancer is flanked by a 10xK enhancer and a PH-2 gene.

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30 kW module · DC-DC (upper) board · sheet 2 of 3 · 1x Vienna PFC cell + 1x 3-ph LLC cell
14 sections · 158 wired clusters · 282 symbols · cross-section links are net labels

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LCSC: calculations/out/bom-30kw.csv